REP-105 · Rules version 18 built and checked: results and what is left to decide

Game	Fruit Market 1.19.1 live; rules version 18 on the working branch (not released)
Report date	2026-11-12 (Asia/Kuala_Lumpur)
Type	Build and results report
Request	The decisions in DEC-086 (the REP-101 proposal) and DEC-088 (Favoured Trade after the flip)
Approval	Building approved (PLAN-079). Publishing: not requested, not done.

The build was made by a helper in a separate work folder, then merged into `ccr-d8bcd65d-gblqqb`. The computer-game figures below are the helper's (`docs/studio/review-17/balance/RULES-18-RESULTS.md`, with the scripts that made them); I have not re-run them. The checks in section 5 are mine, on the merged code.

1. What is built

Item	Built as
$5 top-up	At dawn from day 2, a stall with no fruit at all and under $5 goes to $5. Public, shown on the morning card. Paid before the buying order is worked out
Staked standing orders	The card's old fine is the stake ($6, $8, $7, $12; halved with an Order Book). You need the stake to bid; the winner pays it; a full delivery returns it; anything less loses it. Rivals no longer fill the gap
Shopkeepers	Tip Jar $1 shared, $2 alone. Nest Egg $15. Sharp Ears $2 at 3 to 6 stalls, $1 at 2, once a day, in the evening. Normal computers no longer price as if they were the Haggler
Upgrades	Juice Bar $6, Premium Display $5, Ripening Room $4 (green mangoes $1 less), Order Book 3(+2 a delivered piece, halves the stake, wins tied bids), Handcart $6
Fruit and decks	Green mango $4. Three Bargain customers pay $4 for a blueberry. Short Load and Summer Run rows corrected. Mango Windfall's extra mangoes arrive ripe
Days 1 and 2	No customer wants mango; Mango Craze cannot be day 2's event
Today's purchases	On show for every stall until the next morning, on rival boxes and in the rival pop-up
Rulings	Juice Bar's $1 in the rush only; Premium Display never on a guest; Priority beats Nest Egg on day 3; one Priority an evening; shared win after cash and fruit sold; one favour deck of 12
Nothing to decide	No stocking or pricing turn with no ripe fruit; no evening turn without a choice
Favoured Trade	A new step after the prices flip, only when a holder has a tied price; asked in buying order, in the open
Versions	Rules 18, computer players 6. Unfinished rules 17 games are not resumed; the daily board starts fresh

Also fixed: the day story could count pieces an earlier customer had already taken.

Choices the helper made beyond the brief, for the owner to know: the top-up does not count as earnings in the stats; a lost stake counts as money spent; the mango-customer delay applies only in games with green mangoes; a guest joins a full row and Blueberry Glut makes blueberries $1 cheaper (both tested, below); a computer plays Favoured Trade when it adds $2 or more that day; places are named up to 6th.

2. "Simulate first" verdicts

Item	Verdict	Why
The stake, at the card's old fine	**Kept**	Careful bidding is worth $0.40 to $1.00 a game; wild $1 bids lose about $30. Lower or higher stakes change how many orders find a bidder, not whether bidding pays
+1 apple on five lorries	**Not built**	Apple supply against demand only moves from 0.74 to 0.81 (3 stalls) and the shopkeeper spread gets worse at 4 and 6 stalls
A guest joins a full row	**Kept**	At 5 and 6 stalls a guest event went from costing each stall about $1 to earning about $2
Blueberry Glut $1 cheaper	**Kept**	The event went from costing $0.50 a stall to about even

3. Results against the targets

Shopkeepers: largest distance from a fair share of wins (target: 4 points or less).

Table	Rules 17	Rules 18	Furthest now
2 stalls	8.6	**4.5**	Charmer +4.5
3 stalls	10.4	**3.3**	Nest Egg -3.3
4 stalls	8.6	**4.0**	Charmer +4.0
6 stalls	6.9	**4.6**	Charmer +4.6
3 stalls, Hard	9.3	**3.9**	Nest Egg -3.9
4 stalls, Hard	8.3	**4.2**	Bargain Hunter +4.2

44 of 48 results are inside the target (28 before). The target is **missed, narrowly**: the Charmer is above fair at every table.

Other measures.

Measure	Rules 17	Rules 18
Games with a stall stuck at $0 (2 / 3 / 4 / 6 stalls)	2.1 / 3.6 / 7.4 / 13.2%	0%
Games where a top-up is paid	none	1.3 / 1.4 / 2.2 / 5.4%, about $4.40, never before day 5
Games won by $20 or more	54.7 / 48.5 / 45.3 / 44.8%	52.7 / 43.9 / 41.1 / 39.4%
Standing orders with no bidder (3 stalls)	9.1%	15.8%

Measure	Rules 17	Rules 18
Orders delivered in full (3 stalls)	79.9%	83.4%
Best upgrade against worst (3 stalls)	$10.17 apart	$6.15 apart
Favoured Trade, one free copy	+$0.39 a game	+$2.98 a game
Favoured Trade step comes up	(played blind)	1 to 2 days a game

4. What is left for the owner

Three choices, none urgent; each was tested on a copy or measured, and none is built.

1. **The Charmer starts with 3 favour cards, not 4.** This puts every shopkeeper within 4 points at 2, 3, 4 and 6 stalls with Normal computers (largest distance 2.6 / 3.1 / 1.8 / 3.7). Not yet run with Hard computers. Recommended.
2. **The Ripening Room's $1 discount on green mangoes.** The owner chose it, but it largely undoes the $4 green mango: stalls with the room pay about $3.40 to 3.55, 50*to*904.13 a game). Options: keep it; or drop the discount and keep the faster ripening.
3. **Mango standing orders.** About half find no bidder (41.5 / 53.4 / 46.0% at 3 / 4 / 6 stalls, against 17.6% before), because the $12 stake and the green mangoes must be paid for at once. A $9 stake helps only a little. Options: leave it (an order nobody wants is simply skipped); or look for another fix.

Noted, no decision asked: Free Fruit is the strongest favour (+$4.92 a copy). The Handcart loses a Normal computer $2. Mango Windfall still costs a stall $1.57 on the day. Tourist Bus does nothing at 5 and 6 stalls. One Hard computer among Normal ones now wins 73 / 67 / 60% at 2 / 3 / 4 stalls, because Hard remembers every purchase and Normal only today's. Big Basket is -3.7 at 6 stalls.

5. Checks

Check	Result	Evidence
`npm run verify` on the merged code	PASS	2026-11-12: 518 tests passed, 1 skipped (an old skip); lint, types and build clean. `tests/rules18.test.ts` has 65 tests
Browser tests (`npm run test:e2e`)	PASS	2026-11-12: solo game with no errors; performance budget OK; online match finished with 0 leaks; quick match; reconnect 5 of 5
Balance target (all shopkeepers within 4 points)	FAIL, narrowly	Section 3
The new screens looked at in a browser by a person	NOT RUN	The stake panel, the Favoured Trade question, purchase marks on rival boxes (may clip at 5 and 6 stalls) and the top-up line have unit tests only
Played by people	NOT RUN	Computers do not bluff, never play Scout and barely use Gossip, so open purchases, open bidding and the Favoured Trade step need a playtest
Printed files rebuilt for rules 18	NOT RUN	Text updated in the generators; PDFs wait for the print kit package